Nunna Mokshith

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nunna-mokshith.github.io/mokshith-portfolio

SUMMARY

B.Tech student in Robotics and Artificial Intelligence at Amrita Vishwa Vidyapeetham (Semester 5 | 8.96 CGPA) with active research contributions spanning Physics-Informed Machine Learning, Agentic AI Systems, and Embedded IoT Automation. Author of 2 accepted conference publications — an autonomous vehicle navigation paper at ICRM 2025 and an industrial IoT automation paper at IEEE ICICNIS 2026 and a Finalist (Milestone 3) in the IEEE IES 2026 Generative AI Challenge for physics-guided 3D molecular generation, with 2 further manuscripts under review at IEEE/Springer and Elsevier. Research interests include physics-constrained generative models, multi-agent orchestration architectures (LangGraph, MCP, ReAct), and mission-critical embedded systems design. Developer of Wellsy v2.0, an autonomous AI desktop daemon featuring multi-interface command control, encrypted memory, and tiered risk governance. Certified in MLOps for Generative AI and Responsible AI through Google SkillsBuild. Actively seeking Research & Development internship programmes

Education

08/2024 – Present — **Bachelor of Technology (B.Tech) in Robotics and Artificial Intelligence,** *Amrita Vishwa Vidyapeetham* — Amritapuri
CGPA: 8.9/10.0 | Current Semester: S5

Relevant Coursework: Linear Algebra, Probability & Statistics, Multivariable Calculus, Differential Equations & Numerical Methods, Robot Kinematics, Robot Dynamics, Control Systems, Sensors & Signal Processing, Programming in C/C++/Python, Actuators & Drives, Engineering Mechanics.

07/2022 – 03/2024 — **Higher Secondary Education,** *Narayana E-techno Junior College* — Tirupathi
Board of secondary education - Percentage : 94.7%

07/2021 – 05/2022 — **Secondary School Certificate (SSC),** *Narayana EM High school* — Palamaner
Board of Secondary Education, Andhra Pradesh (BSEAP) - Percentage: 93.67%

Research & Publications

07/2026 — **Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation,** *2nd International Conference on Robotics and Mechatronics (ICRM 2026)*
Submitted (Paper ID: 918). IEEE IES 2026 Generative AI Challenge — Finalist (Milestone 3). Proposed a dual-head Graph Neural Network (GNN) with Physics-Informed Neural Operator (PINO) auxiliary loss for thermodynamic-constrained diffusion on QM9 molecular graphs. Achieved 89.5% structural validity and 76.5% Lipinski compliance via a 3-step RDKit validation pipeline.

07/2026 — **Domain-Aware Multi-Modal Industrial Diagnostics: A Framework Integrating Mechanical Heuristics with Ensemble Learning and Conformal Uncertainty,** *2nd International Conference on Robotics and Mechatronics (ICRM 2026) — IEEE*
Submitted (Paper ID: 879). Developed a Physics-Informed Machine Learning (PIML) predictive maintenance framework using a Dual-Channel Kalman Filter across three sensor modalities. Integrated Adaptive Conformal Prediction for uncertainty quantification under sensor drift. Achieved 69.6% recall on highly imbalanced multi-hazard datasets, validating $272,600 median maintenance savings across 2,000 devices.

06/2026 — **Autonomous Dual-Mode Failover Architecture for Multi-Device IoT Switching using 4G LTE and Localized HTTP Gateways,** *ICICNIS 2026 — IEEE | T. John Institute of Technology, Bengaluru*
Accepted & Recommended for Publication. Engineered a bimodal control system integrating ESP8266 microcontroller with A7670C 4G LTE module. Developed a Carrier Signal Quality (CSQ) thresholding algorithm with autonomous HTTP Soft Access Point failover ensuring zero-downtime command integrity for industrial relay channels. Conference: Nov 23–25, 2026.

06/2026 — **Performance Evaluation and Desalination Efficiency Analysis of Low-Cost Parabolic Dish Solar Desalination System for Tropical Coastal Applications,** *Elsevier — Next Energy Journal*
Under Review (Manuscript ID: NXENER-D-26-01950). Conducted empirical energy balance modeling and thermal efficiency analysis for a parabolic dish solar desalination system, evaluating yield improvements across varying thermal flux levels for tropical coastal deployment.

11/2025 **Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity,** *International Conference on Robotics and Mechatronics (ICRM)*
Status: PUBLISHED & PRESENTED | International Conference on Robotics and Mechatronics (ICRM 2025)
Architected a hardware-in-the-loop Raspberry Pi platform emulating Level 3–4 autonomous driving capabilities. Deployed a multi-sensor array and computer vision pipelines for real-time obstacle perception, bridging classical control theory with machine learning for complex environmental navigation.

Projects

08/2026 **GestureSense — OpenCV & MediaPipe Cybernetic Air Interface**
Engineered a real-time spatial air interface using MediaPipe 21-landmark hand tracking at 30+ FPS. Developed touchless air mouse control algorithms with a custom gesture trainer for real-time cursor navigation and system controls. Integrated a spatial air piano synthesizer mapping 3D gesture coordinate matrices to audio synthesis loops. Stack: Python, OpenCV, MediaPipe, NumPy.

07/2026 **Wellsy v2.0 — Multi-Interface Autonomous AI System Daemon**
Engineered a persistent background system daemon (pystray) featuring a multi-interface AI command center: a dark HUD web dashboard (FastAPI + WebSockets), interactive terminal CLI, and a persistent scheduler engine. Implemented AES-256 Fernet encrypted SQLite memory with passive file metadata scanning. Designed a 4-tier risk governance system (SAFE → LOW_RISK → HIGH_RISK → DANGER) with async WebSocket confirmation gates intercepting system/git actions for manual UI approval. Deployed a custom Model Context Protocol (MCP) server for standardized agent tool execution. Stack: Python, FastAPI, WebSockets, pystray, LangGraph, SQLite, AES-256.

06/2026 **Wellsy v1.0 — Agentic AI Personal Assistant**
Engineered a local agentic AI personal assistant using LangGraph ReAct loop architecture with persistent multi-session memory. Integrated tool-use modules spanning calendar management, system controls, Spotify automation, WhatsApp dispatch, and email orchestration via Model Context Protocol (MCP). Enabled conversational interaction with real-world digital systems using autonomous decision-making and action chaining. Stack: Python, LangGraph, LangChain, MCP, FastAPI, Electron.js.

05/2026 **Domain-Aware PIML Predictive Maintenance Framework**
Developed a Physics-Informed Machine Learning (PIML) framework for industrial predictive maintenance integrating Dual-Channel Kalman Filter across three sensor modalities with Adaptive Conformal Prediction for uncertainty quantification. Achieved 69.6% recall on highly imbalanced multi-hazard datasets. Monte Carlo simulation validated $272,600 median maintenance savings across 2,000 devices. Submitted as research paper to ICRM 2026 (Paper ID: 879). Stack: Python, Scikit-learn, Kalman Filter, SHAP, Monte Carlo Simulation.

05/2026 **Evolution Edge — Self-Evolving Neural Bridge**
Engineered a hybrid AI inference pipeline utilizing ROCm 6.x and ONNX Runtime to orchestrate low-latency inference on Ryzen AI NPUs with confidence-based symbolic escalation to AMD Instinct MI300X GPU clusters. Implemented persistent on-device learning via knowledge distillation and PEFT-driven weight updates for continuous model adaptation without full retraining. Stack: Python, ROCm 6.x, ONNX Runtime, PyTorch, PEFT. Developed during the global AMD Developer Hackathon (lablab.ai) using Antigravity & WebGPU (Cert ID: CMQHZXVHS00SPS601R3ZN38H4).

04/2026 **Physics-Guided Diffusion with Thermodynamic Awareness — 3D Molecular Generation Framework**
Engineered an end-to-end generative AI framework for de novo catalyst discovery, addressing the high computational cost of Density Functional Theory (DFT) calculations. Implemented a dual-head Graph Neural Network (GNN) with Physics-Informed Neural Operator (PINO) auxiliary loss for thermodynamic-constrained diffusion graph generation on QM9 molecular datasets. Integrated Gibbs free-energy loss for structural stability enforcement, a 3-step RDKit validation cascade, and 3D molecular visualization pipeline. Achieved 89.5% structural validity (+15.5% over baseline) and 76.5% Lipinski drug-likeness compliance. IEEE IES 2026 Generative AI Challenge — Finalist (Milestone 3). Submitted to ICRM 2026 (Paper ID: 918). Stack: Python, PyTorch, PyTorch Geometric, RDKit, QM9, PINO.

11/2025 **Multi-Parameter Health Vitals Monitoring & Real-Time IoT Data Streamer**
Architected an embedded IoT health-monitoring node using an ESP8266 (NodeMCU) microcontroller. Interfaced a MAX30100 optical pulse oximetry sensor via I2C for continuous SpO2 and Heart Rate (BPM) acquisition, coupled with a DHT11 sensor for ambient clinical temperature and humidity tracking. Developed firmware in Embedded C++ implementing non-blocking sampling routines and integrated Blynk IoT cloud protocols for real-time mobile telemetry visualization and alert streaming. Stack: Embedded C++, ESP8266 NodeMCU, I2C Protocol, MAX30100, DHT11, Blynk IoT Cloud.

12/2025 — **Hybrid GSM & Wi-Fi Smart Switch — Industrial IoT Automation System**
Designed and fabricated a mission-critical dual-mode industrial automation switch coupling ESP8266 NodeMCU with A7670C GSM/LTE module. Implemented Carrier Signal Quality (CSQ) thresholding with autonomous 4G-to-Wi-Fi failover ensuring 100% operational uptime. Engineered fail-safe asynchronous SMS interrupt parsing for zero-internet remote hardware reset. Custom PCB schematic designed for field deployment. Published at ICICNIS 2026 (IEEE). Supervised by Prof. Sumesh K.P. Stack: C++, Arduino, ESP8266, A7670C, GSM AT Commands.

HACKATHONS & COMPETITIONS

07/2026 — **Amrita Hack Horizon Hackathon — Fin-Analyst-Agent,** *Nitrostack & WeKan Enterprises*
Built 'Fin-Analyst-Agent', a Model Context Protocol (MCP) financial analysis and risk engine on NitroStack. Integrated Yahoo Finance API for real-time stock profiling, quarterly revenue/EBIT metric extraction with QoQ growth analysis, 50/200-day moving average momentum scoring, and automated AI buy/hold/sell investment recommendations. Codebase & system implementation verified by CEOs Abhishek Pandit (NitroStack) & Pablo Jiménez Godoy (WeKan Enterprises). Stack: TypeScript, NitroStack MCP SDK, Yahoo Finance API, Next.js.

05/2026 — **AMD Developer Hackathon (Lablab.ai),** *lablab.ai (Part of NativelyAI) × AMD*
Successfully built and submitted 'Evolution Edge', a hybrid AI neural bridge based on Antigravity and WebGPU during the 7-day global AMD Developer Hackathon. Orchestrated low-latency edge inference on Ryzen AI NPUs with confidence-driven escalation to AMD Instinct GPU clusters. Earned official Certificate of Completion (Cert ID: CMQHZXVHS00SPS601R3ZN38H4). Stack: Antigravity, WebGPU, ROCm, ONNX Runtime, PyTorch, PEFT.

02/2026 — **Guidewire DEVTrails University Hackathon — AI Parametric Insurance Engine,**
Guidewire Software × National Insurance Academy (NIA) × EY India
Engineered an AI-powered parametric income insurance engine protecting gig delivery workers from forced downtime due to severe weather. Built zone-based geo-mapping with 15-minute hyper-local weather API polling, a dynamic AI risk-scoring engine combining historical trends with real-time feeds, and an automated parametric payout pipeline triggering digital wallet payouts in under 60 seconds with zero paperwork. Competed nationwide against 18,000+ students across 30 universities.

Skills

Embedded & Robotics
ESP8266-NodeMCU, Arduino UNO, GSM/LTE (A7670C), Wi-Fi, I2C Protocol, Control Systems, Sensors (MAX30100, DHT11, Ultrasonic)

Programming
Python, C, C++, Embedded C++, HTML/CSS

Design & Tools
AutoCAD,Git/GitHub, Geany.

AI & Agentic Systems Architecture
Agentic workflows Design and Management, LangGraph, RAG Pipelines, FastAPI, WebSockets. Antigravity, Model Context Protocol (MCP)

Soft Skills
Problem Solving
Leadership Skills and Teamwork
Project Management and Research
Adaptability & Crisis Management

Machine Learning
Physics-Informed Machine Learning (PIML), Graph Neural Networks (GNN), Predictive Maintenance, Exploratory Data Analysis (EDA), Data Normalization, SMOTE, SHAP (Explainable AI)

Web & Software Frameworks
React, Next.js, Electron.js, Node.js, WebGPU, ONNX Runtime

Certificates

MLOps for Generative AI
Google SkillsBuild | July 2026

Responsible AI for Developers: Fairness & Bias
Google SkillsBuild | July 2026

Guidewire DEVTrails University Hackathon — Certificate of Participation / Achievement
Guidewire Software × National Insurance Academy (NIA) × EY India | March 2026

Quantum Computing: IBM Quantum Qiskit
November 2025

MLOps with Agent Platform: Model Evaluation
Google SkillsBuild | July 2026

Data Science Course - Mastering the Fundamentals
Scaler| April 2026

Google Cloud Computing Foundations: Cloud Computing Fundamentals
Google skills | June 2026

AMD Hackathon Certificate
lablab.ai | (Part of NativelyAI) × AMD | May 2026

Leadership & Co-Curricular Activities

Field Hockey Team Captain & Center Player, *Amrita Vishwa Vidyapeetham Athletics*
Captained and led a 20-member field hockey squad across high-stakes inter-college competitive tournaments. Responsible for real-time tactical positioning, squad rotation, on-field split-second decision-making, and team coordination under intense pressure. Fostered high situational confidence, strategic communication, and rapid problem-solving skills directly applicable to software architecture and engineering team management.